Rabin Karmakar

Software Engineer | Full Stack

Email: rabinkarmakar947@gmail.com | Phone: +91 8967873860 | Bangalore, India
LinkedIn: linkedin.com/in/rabin-642894219 | GitHub: github.com/rabin97 | Portfolio: profile-blue-one.vercel.app

Professional Summary

Full stack Software Engineer with 3.5+ years of hands-on experience building production **web applications**, **REST APIs**, and **microservices** across the full **SDLC**. Builds intuitive frontends with **React.js**, **Next.js**, **TypeScript**, **JavaScript (ES6+)**, **HTML** and **CSS**, and backend services in **Python** and **Node.js (NestJS)** backed by **PostgreSQL** and **MongoDB**. Experienced with **CI/CD**, **Docker**, and **Redis**; writes clean, testable code, drives **code reviews** in **Agile** teams, supports production releases, and has led teams of 10+ developers. Exposure to **AI/ML** via M.Tech research in deep learning.

Professional Experience

Software Engineer *Sept 2025 – Present*
Digitap.ai

- **Owned frontend architecture** for a production-grade application built with **React.js** and **TypeScript**, driving end-to-end feature delivery and technical decisions in collaboration with backend and product teams
- Designed and implemented a **hybrid AES–RSA encryption system** with **session-based AES key exchange**, securing all frontend-to-backend API communication
- Led **code reviews**, **coding standards**, and **Agile sprint execution**, and mentored junior and mid-level developers, improving code quality and release predictability
- Refactored legacy codebases using **OOP** and **SOLID principles** into clean, testable, modular components, improving scalability and long-term maintainability

Full Stack Developer *Dec 2023 – Sept 2025*
Explorogent International Services Pvt. Ltd. (Stackkaroo)

- Designed a **microservices**-based backend in **Python** and **Node.js (NestJS)** with **Prisma ORM** and **PostgreSQL**, exposing **REST APIs** consumed by a **React.js (Next.js)** + Redux Toolkit frontend, supporting complex multi-role workflows and high reliability
- Implemented **CI/CD** pipelines with **GitHub Actions**, reducing deployment time by 40% and improving release consistency
- Led a 10-member cross-functional team, owning system architecture, requirements scoping, sprint planning, and delivery from design through production launch and post-release support
- Architected and shipped CryoFlow, a large-scale logistics and pharmaceutical delivery platform with real-time tracking and compliance systems, delivered to customers
- Built secure, high-performance services with **JWT authentication**, **Role-Based Access Control**, **AWS S3** integration, and **Redis** caching

MERN Stack Developer *Oct 2022 – Dec 2023*
Explorogent International Services Pvt. Ltd. (Stackkaroo)

- Designed and implemented high-performance **RESTful APIs** with JWT authentication (**Access** and **Refresh Tokens**), handling substantial monthly traffic
- Optimized **MongoDB** aggregation pipelines, reducing processing time by 28%, and streamlined **React** app architecture with Redux Toolkit

Technical Skills

Languages:	**Python**, **JavaScript (ES6+)**, **TypeScript**, **SQL**, **HTML**, **CSS**
Frontend:	**React.js**, **Next.js**, Redux Toolkit, RTK Query, Tailwind CSS
Backend:	**REST API Design**, **Node.js**, **NestJS**, Flask, Express, WebSockets
Architecture:	**Microservices**, **OOP**, **SOLID Principles**, Design Patterns, RBAC, Feature Flags, JWT Auth, AES–RSA Encryption
Databases:	**PostgreSQL**, **MongoDB**, Redis, Prisma ORM, Mongoose
DevOps & Tools:	**CI/CD** (GitHub Actions), **Docker**, AWS (S3, RDS), **Git**, **GitHub**, Postman, ESLint, Husky
AI/ML & Data:	Deep Learning (Swin Transformers), pandas, NumPy, Matplotlib
Practices:	**Agile/Scrum**, **SDLC**, **Code Reviews**, Clean & Testable Code, Documentation, Mentoring, 250+ **DSA** problems solved

Key Projects

CryoFlow – Cold/Non-Cold Chain Logistics Platform *2024–2025*

- Built a full stack pharmaceutical logistics platform (**NestJS**, **Next.js 15**, **PostgreSQL**) with core services for order, inventory and delivery management, **RBAC (7 roles)**, real-time tracking, temperature-compliance monitoring, **Redis** caching, **AWS S3** storage and automated PDF generation

Tierloom – Creator-Commerce Platform (Independent) *Ongoing*

- Building a creator-commerce platform with **Next.js** and **NestJS**: free vs. premium plan architecture with **feature flags** and a reusable guard/decorator pattern, plus a production-readiness review of auth (access/refresh tokens, race-condition fixes, security headers, secure session cookies)

Financial Analyst AI – Automated Stock & Crypto Analysis Engine *2025*

- Built a **Python** engine that ingests market data and news feeds, computes volatility, historical VaR and technical indicators (RSI, MACD, Bollinger Bands) with **pandas**/**NumPy**, and drives automated BUY/HOLD/SELL and trade-planning logic through a web UI

Education

B.Tech in Computer Science & Engineering *2020 – 2024*
Bankura Unnayani Institute of Engineering Bankura, India
CGPA: 8.66/10